

Bicycle Traffic Flow Prediction

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October 2025

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Using Graph Neural Networks for Predicting Campus Cycling Dynamics

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Date	October 2025

Preface

This thesis has been written for the Bachelor programme in Civil Engineering at Delft University of Technology. The project was carried out in the Department of Transport & Planning during the first quarter of the academic year 2025–2026. With this study I complete the Bachelor End Project (BEP).

During my bachelor, I found a big interest in infrastructure and traffic flows and their corresponding courses. This project allowed me to combine that interest with my minor in Artificial Intelligence, exploring how modern data driven approaches can be applied within the field of transport engineering. Engaging with both fields has provided valuable insight into the potential of AI to enhance sustainability and innovation in mobility planning.

Acknowledgements. I would like to express my sincere gratitude to my supervisors, Prof. dr. R.M.P. Goverde and Prof. dr. M.P. Hagenzieker, from the Department of Transport & Planning, for their guidance, constructive feedback, and support throughout this project. I also like to thank Mingze Gong, Dr.ir. W. Daamen and Drs. ing. I.L. Oostlander-Çetin for providing access to the bicycle traffic data and helping me refine the research topic. I further wish to thank my fellow students and friends for their valuable discussions, feedback and encouragement during the process.

Delft, October 2025

Raf van de Kar

Summary

Cycling is a defining element of Dutch mobility and plays a crucial role in creating sustainable and accessible cities. The TU Delft campus serves as a clear example of this, with thousands of daily bicycle trips linking housing, faculties, and public transport. To ensure safe and efficient operation of this infrastructure, accurate forecasts of bicycle traffic flows are needed to support maintenance, capacity management, and planning. Traditional statistical models can estimate trends but struggle to capture the spatial relationships between intersections. With recent advances in artificial intelligence, Graph Neural Networks (GNNs) have emerged as a promising method for learning such spatial–temporal dependencies.

This study investigates to what extent bicycle traffic flow on the TU Delft campus can be predicted and what additional value GNNs provide over classical approaches. Data from January to May 2025 were obtained from campus sensors placed by TU Delft Campus Real Estate & Facility Management (CREFM). The data was combined with contextual features such as temperature, rainfall, wind, and calendar variables. After preprocessing and network construction, three predictive models were developed and compared: a historical average baseline, a linear regression model, and a spatio-temporal GNN (ST-GNN). All models were trained and evaluated using a 70-15-15 train-validation-test split, with mean absolute error (MAE) as the primary performance metric.

The results show that the ST-GNN achieves a substantial improvement in prediction accuracy, reducing MAE by 30.1% relative to the baseline. This shows that spatial and temporal dependencies within the network have significant predictive value. While the linear regression model benefitted from contextual data, it lacked the ability to propagate flow information between connected intersections. In comparison, the ST-GNN effectively modelled these relationships and provided smoother and more realistic forecasts. Contextual factors such as temperature, rain, and weekend indicators further enhanced performance, though their influence was smaller compared to spatial network effects.

The research concludes that bicycle traffic flows on the TU Delft campus can be predicted with accuracy comparable to that reported for vehicular traffic in previous studies. Nevertheless, limitations such as the short four month dataset, limited sensor coverage, and computational constraints restrict generalisability. Future work should extend the dataset to a full year coverage, install additional sensors at key intersections, and explore larger or heavier model architectures. These improvements would enable more precise, generalisable, and interpretable prediction models to support data driven bicycle infrastructure management on campus and beyond.

Contents

Preface	i
Summary	ii
1 Introduction	1
1.1 Motivation and problem analysis	1
1.2 Objectives and scope	2
1.3 Methodology overview	2
1.4 Reading guide	3
2 Methodology	4
2.1 Literature Review	4
2.2 Data Collection and Preprocessing	4
2.3 Model Development	5
2.4 Performance Evaluation	5
2.5 Interpretation and Visualization	5
2.6 Reproducibility and Limitations	5
3 Literature review	7
3.1 Classical and statistical models	7
3.2 Machine and Deep Learning Models	7
3.3 GNNs in traffic forecasting	8
3.4 Challenges and relevance to bicycle traffic prediction	9
4 Data and preprocessing	10
4.1 Dataset	10
4.2 Additional features	11
4.3 Train-Test split	11
4.4 Campus Network	12
5 Model development	13
5.1 Models	13
5.2 Hyper-parameter grid search	14
5.3 Training	15
6 Results and performance evaluation	16
6.1 Network representation	16
6.2 Role of contextual features	17
6.3 Model comparison	17
6.4 Comparison with literature	19

6.5	Visualisation of the results	20
7	Discussion	21
7.1	interpretation of the results	21
7.2	limitations/assumptions	21
8	Conclusions and recommendations	23
8.1	Conclusion	23
8.2	Recommendations	23
A	Word explanations	25
B	Extended Figures	27

Chapter 1

Introduction

This chapter introduces the research topic and outlines the motivation, problem context, and research objectives. It analyses the challenges of predicting bicycle traffic flow, formulates the main research question and sub-questions, and provides an overview of the methodological approach used to address them. The chapter concludes with a reading guide explaining the report structure.

1.1 Motivation and problem analysis

Cycling is a core part of Dutch culture and its urban infrastructure, the usage of bikes reduces the reliance on cars and other fuel powered vehicles while needing less infrastructure. Therefore the bicycle can be a good tool in the fight against CO₂ emissions and the current nitrogen crisis. The TU Delft campus is a prime example of how bicycles can be the main form of transportation by connecting housing, faculties and train stations efficiently. However the bicycle infrastructure has to overcome challenges like capacity during peak hours, safety and be able to adapt to construction work and other disruptions.

To help with planning for these peak hours and disruptions a accurate prediction of bicycle traffic flows is critical to maintain accessibility. Accurate predictions give valuable insight into how maintenance and road closures can affect bicycle traffic. The problem now is that accurate predictions are difficult to make. This is because next to the infrastructure itself, cycling is also impacted by other external factors such as rain, slipperiness and time of day more than some other transportation methods. Modern AI models are well suited to take into account all kinds of factors and have improved accuracy of predictions in many different fields, therefore a interesting opportunity to use these AI models on bicycle traffic flow presents itself.

Studying bicycle traffic was traditionally done with regression analysis or other statistical models such as Kalman filter model. While these models can provide working and usable results, these are not able to take into account the spatial structure of the network. With the emergence of AI a new method has emerged within transportation research(R. Liu et al., 2025).

One of the promising AI models in this area is the Graph Neural Network, hereafter called GNN for short, which is able to take in account the spatial structure of the network together with other factors that could be of influence. While GNN models have been used successfully to predict car traffic, applying this to bicycle traffic seems not fully studied yet.

Currently we find two relevant problems within bicycle traffic flow predictions: The limited accuracy of traditional statistical prediction models and being able to use the external factors in bicycle traffic flow prediction.

1.2 Objectives and scope

To address the problems with bicycle flow prediction this study aims to create a prediction with a GNN model which can improve on existing prediction models and incorporate external factors into its predictions. To test this within the scope of this 8 week thesis we will only focus on a GNN model, linear regression model and a historical average baseline. As a large citywide or even national model of the bicycle network is unobtainable within the scope, the geographical scope will be focused on the TU Delft campus. The data used covers January to May 2025 to reduce training times of the model. And only certain external factors such as weather will be included.

In short this study will analyse how accurate bicycle traffic can be predicted and what the difference in accuracy are between traditional methods and a GNN model. Specifically this study will try to answer the question:

To what extent can bicycle traffic flow on the TU Delft campus be predicted and what value can Graph Neural Networks provide over other approaches?

To answer this question we will partition it into four sub questions to achieve the answer of the main research question:

- How should the TU Delft campus bicycle network be represented as a graph for predictive modeling?
- What use of contextual factors(rain, exam weeks) can improve Graph Neural Network predictions?
- How do predictions differ between a Graph Neural Network and a simpler model of linear regression?
- How do the prediction results for a Graph Neural Network compare to their performance in other transport as reported in literature?

1.3 Methodology overview

The study follows a structured approach. First, a review of scientific literature is conducted to position the study within existing work and to identify relevant methodological insights. Second, bicycle traffic data collected by TU Delft Campus Real Estate & Facility Management (CREFM) is pre-processed and combined with contextual datasets such as weather and academic calendars. Third, three models are developed: a GNN, a regression model, and a baseline. These were trained and evaluated on the same dataset using standard error metrics such as Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE). Finally, the results are compared across models and interpreted with reference to the literature.

This stepwise approach enables systematic evaluation of both the predictive performance of GNNs and their relative advantages over established methods.

1.4 Reading guide

The remainder of this report is structured as follows. Chapter 2 describes the methodology in detail, outlining how the research questions are answered and how the models are constructed. Chapter 3 presents the related work, including a review of both traditional and AI-based traffic prediction methods, with particular attention to missing attention in bicycle traffic research. Chapter 4 introduces the data sources, preprocessing steps, and graph representation of the campus. Chapter 5 presents the development of the predictive models. Chapter 6 reports on the results and performance evaluation, followed by Chapter 7, which provides a critical discussion of the findings, limitations and implications. Finally, Chapter 8 concludes the thesis with answers to the research questions and recommendations for further research and practice.

Chapter 2

Methodology

This chapter describes the methodological approach taken to answer the research question: “To what extent can bicycle traffic flow on the TU Delft campus be predicted and what value do Graph Neural Networks provide over other approaches?” The structure of this chapter follows how each of the subquestions defined in Chapter 1 Introduction will be researched and answered. Each step needed to answer the subquestions as stated here will be followed during the study.

2.1 Literature Review

This research study begins with the review of scientific literature on different transport modelling methods. This is needed to show the relevancy of this research topic and are used to answer the fourth subquestion, which asks how the performance of GNN’s for bicycle traffic compare to their use in other forms of transport. The review of scientific literature was used to show the connections and differences between using GNN’s for bicycle traffic and other traffic forms. Platforms such as IEEE Xplore, TU Delft repository, Scopus and Google scholar were used to search for relevant literature, search terms such as graph neural network, traffic prediction, bicycle traffic prediction and traffic modelling. This literature was referenced in APA style in line with the TU Delft standards (TU Delft, Faculty of Civil Engineering and Geosciences, 2025).

2.2 Data Collection and Preprocessing

The research is based around bicycle traffic data provided by TU Delft Campus Real Estate & Facility Management (CREFM). This data consists of sensor counts of bicycles entering and exiting different intersections on the TU Delft campus from January till may 2025. It was collected using cameras to detect whenever a cyclist crosses a virtual line into or out of the intersection. This data together with the topological layout of the campus was used to address the first subquestion, how the TU Delft campus network should be represented as a graph for use in the predictive model and for visualisation. Other than bicycle traffic counts, data such as weather, day of the week and if it is currently exam week or not was used to give contextual data for the GNN model to interpret. The best combination of contextual data is then checked though grid search. This corresponds to answering subquestion two, in which contextual external factors was evaluated. Weather data from Januari till May 2025 was taken from the Davis weather station on top of the EWI building (Schleiss et al., 2025).

When all data is collected, preprocessing handles missing entries, reduces noise, combines external data and splits the dataset into train, validation and test subsets. This last step is needed to make sure the AI model is tested correctly and has not seen the data that it will try to predict, which is needed to ensure the reproducibility and reliability of this study (Medar et al., 2017).

2.3 Model Development

To evaluate to what extent bicycle traffic can be predicted a combination of a GNN model, a regression model and a baseline model in which the historical average was used as a benchmark, was programmed to be able to reference them to each other and provide interpretable reference points. These models are necessary to answer subquestion three, how the predictions differ between the GNN, regression and historical average.

The models were programmed in Python, using the PyTorch package (Paszke et al., 2019) to develop the spacial temporal graph neural network. This model was designed to take advantage of both the layout of the network and the temporal component (L. Liu et al., 2023). For example if one node has increased traffic, the GNN model is able to use the network connectivity to understand some travel time later connected nodes will also have increased traffic.

While hyperparameter optimisation was not included in the initial methodological design, a grid search through key parameter combinations was performed to enhance the GNN's predictive capacity and generalisability.

2.4 Performance Evaluation

To evaluate to what extent bicycle traffic can be predicted, a combination of Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE) were computed. This gives a direct comparison between the three models with which subquestion three can be answered, how the predictions between the GNN, regression and historical average differ from each other by quantifying the results. These errors can also be used to refine subquestion two in which we test how external factors like weather and weekdays impact the predictive capabilities.

2.5 Interpretation and Visualization

The interpretation of the prediction results helps with a better understanding in campus mobility and for future mobility planning. Predictions are be visualised on a network of intersections as nodes and streets as lines, in which line-color together with numbers show the traffic intensities. This way a quick graphical overview can be given which is easy to interpret. The RMSE and MAE are used to interpret how well predictions can be made and answer the research question.

2.6 Reproducibility and Limitations

For reproducibility and transparency all scripts, parameters and model architectures will be made available in a Github repository (github.com/RvdKar/Bicycle_Traffic_Flow_GNN) (van de Kar, 2025). This ensures the study is reproducible and adheres to the scientific criterion of the TU Delft BEP (TU Delft, Faculty of Civil Engineering and Geosciences, 2025). Limitations like data availability, potential over- or under-fitting of the model are analysed and addressed. These potential limitations are used to address subquestion four, in which this study is compared to relevant literature.

Large Language Models (LLMs) were used to support both the writing and coding processes of this study. ChatGPT 5 and Overleaf's integrated AI tool Writefull were used to rephrase sentences, suggest alternative wording, and improve the structure of paragraphs. In addition, ChatGPT 5 was used to generate small code snippets and provide explanations for errors encountered in Python and LaTeX. All conceptual work, analyses, interpretations, and final code implementations were independently verified and remain the responsibility of the author.

Chapter 3

Literature review

This chapter reviews existing scientific literature on traffic flow prediction, highlighting the evolution from classical statistical approaches to modern deep learning and graph-based models. It discusses relevant findings, identifies research gaps regarding bicycle flow prediction, and positions this study within the broader field of transport modelling.

3.1 Classical and statistical models

Within the long history of transport literature, traffic flow prediction has been an important subject. Models based on time series or statistical models such as ARIMA, Kalman filters and regression formed the foundation of these prediction models. These models treat sensor locations mostly independently and assume the traffic flow follows a mostly linear outcome over time.(R. Liu et al., 2025)

Although these limitations are significant, these methods do have advantages in modest data requirements, reduced computational cost and better interpretability. But due to their simplicity and linear limitations, abrupt changes due to weather or other external factors are difficult to account for. These models also struggle to make use of a network structure and understand how changes at one location influence other nodes in the network.(R. Liu et al., 2025)

Multiple studies have documented the strengths and weaknesses of these classical models against modern models. In the study Liu et al. (R. Liu et al., 2025) methods are categorised into statistical, machine learning and deep learning models. The authors argue that classical methods can still have value when used in cases of low data or as a baseline comparison, but are outperformed when more data is available. Similarly Kashyap et al. (Kashyap et al., 2022) come to the conclusion that deep learning methods are well suited for traffic flow prediction and can overcome many of the limitations that statistical models have, mainly the ability to model complex non-linear relations.

In bicycle traffic flow predictions, classical methods are still common, this could be because of lower data count, and more complex spatial interactions. Bicyclists can stop and park nearly anywhere and a simple walking path could provide a connection between two nodes. This makes the spatial interactions difficult to encapsulate.

3.2 Machine and Deep Learning Models

As traffic datasets became larger, richer and more available, machine learning models like support vector regression, random forest and gradient boosting began to see more use. These models are able to use non-linear relations between the data features (eg. Weather, schedules), which leads them to outperform linear models in many situations.

Even more recently deep learning models such as Recurrent Neural Networks (RNNs), Long Short Term Memory Networks (LSTMs) and Gated Recurrent Units (GRUs) are used widely to predict traffic time-series. These models can capture temporal dependencies and patterns across multiple timestamps. Fawaz et al. (Ismail Fawaz et al., 2019) review different deep learning techniques for modelling time series. They summarise the capabilities and challenges of each technique, which can provide in depth explanations out of the scope of this study.

However these Models have a common limitation in that they view each sensor independently from each other and do not use the spatial connectivity of a traffic network. Knowing a change in flow at one location will propagate through the network, can very well be used to enhance predictions at each location. The flow at each node is very much dependent on the flow at the upstream nodes. To make use of this, an approach which uses both temporal and spatial modelling techniques is needed.

Several studies in vehicle traffic predictions incorporate these spatio-temporal techniques, this can be done by combining Convolutional Neural Networks (CNNs) or GNNs with temporal models. Lu et al. (L. Liu et al., 2023) for example, propose using a LSTM with GNN layers to model the spatio-temporal flows. The trend in traffic flow forecasting is moving towards these hybrid deep learning networks, which use both dependencies in space and time.(Cui et al., 2019)

3.3 GNNs in traffic forecasting

Graph Neural Networks (GNNs) are deep learning architectures which make use of graph structured data. For a transport network this will be seen as nodes representing intersections connected by edges representing the bicycle paths between them. These GNNs propagate information from node to node enabling the model to learn how changes at one node impact surrounding nodes.

A number of studies have looked at the foundations and applications of GNNs in general (beyond traffic flows), most notably, Wu et al. (Wu et al., 2021) give a detailed survey of GNN architectures, their limitations and futures for further explanation. In traffic flow forecasting mostly a hybrid architecture of both spatial and temporal layers is used. These Spatial-temporal GNNs (ST-GNNs for short) use both properties of temporal layers and graph layers to achieve the best results.

Some GNN applications in traffic prediction literature include:

- **Yu et al.** propose a Spatial-Temporal Graph Diffusion Network (STGDN) for traffic flow forecasting, with which they show that diffusion-based graph operations can better capture propagation across network links. (L. Liu et al., 2023)
- **Chai et al.** use a graph neural network for bike sharing prediction at the station level. They use multiple inter-station graphs which are then fused to apply convolution layers. Their model achieved a reduction of 25.1% and 17.0% in prediction error compared to their baseline models respectively. (Chai et al., 2018)

3.4 Challenges and relevance to bicycle traffic prediction

The literature on transport flow forecasting with GNNs is constantly growing, however literature also shows using these methods on bicycle flows presents challenges. These explain why this is not yet widely applied in the bicycle flow forecasting and requires more research to be done. Challenges that using GNN in bicycle flow forecasting face are:

- **Irregular sensor coverage.** The coverage of urban networks is often not complete and sparse. This results in difficult to train GNN's as many links may lack direct measurements, which results in weak propagation of spatial dependencies.
- **Differences in network topology.** Bicycle networks often have more flexible routes, shortcuts, informal paths and use of pedestrian paths. This makes the assumptions of the network connectivity difficult and might not hold up steadily.
- **Noisy data.** Bicycle counts are more volatile due to the short travel duration combined with large external impacts such as weather. Bicyclists can take shelter from rain or choose other transportation. In comparison to cars, bicyclists have shorter travel time and distance so schedules of local schools, theatres, train stations are more impactful. Because of this there are heavier short term fluctuations, which makes it more difficult to make a well generalised model.
- **Contextual features.** Bicycle flow is more strongly affected by external variables such as weather, daylight and because of the local nature, schedules and exam weeks. Integrating these contextual features into GNNs is less explored. Although hybrid models which could use these input features have seen more research done recently.

Chapter 4

Data and preprocessing

This chapter presents the datasets and preprocessing steps used for model training and evaluation. It explains how the bicycle count data and contextual variables were obtained, cleaned, and combined, and how the TU Delft campus network was represented as a graph. These preparations form the foundation for the model development described in the next chapter.

4.1 Dataset

The dataset provided by TU Delft Campus Real Estate & Facility Management (CREFM) contains the bicycle counts in both directions over a virtual line in buckets of one minute. The sensors are strategically placed around the campus. The exact sensor locations used and available are found in this study's Github (van de Kar, 2025), not all of these sensors are usable and valuable in this study so a selection is made. To get a better understanding of how the measurements are made the sensors TUD_SBX01 and TUD_SBX07Multi are depicted in Figure 4.1. The green lines here give us our measurement in the form of a count_in and count_out based on the amount of people crossing the lines in their respective direction.

Before this data can be used for the models, first several preprocessing steps are required. First, the data was cleaned of missing entries and where necessary interpolated linearly to fill small gaps. Some sensors reported twice during the same bucket which was rectified by taking the latest data from each bucket.

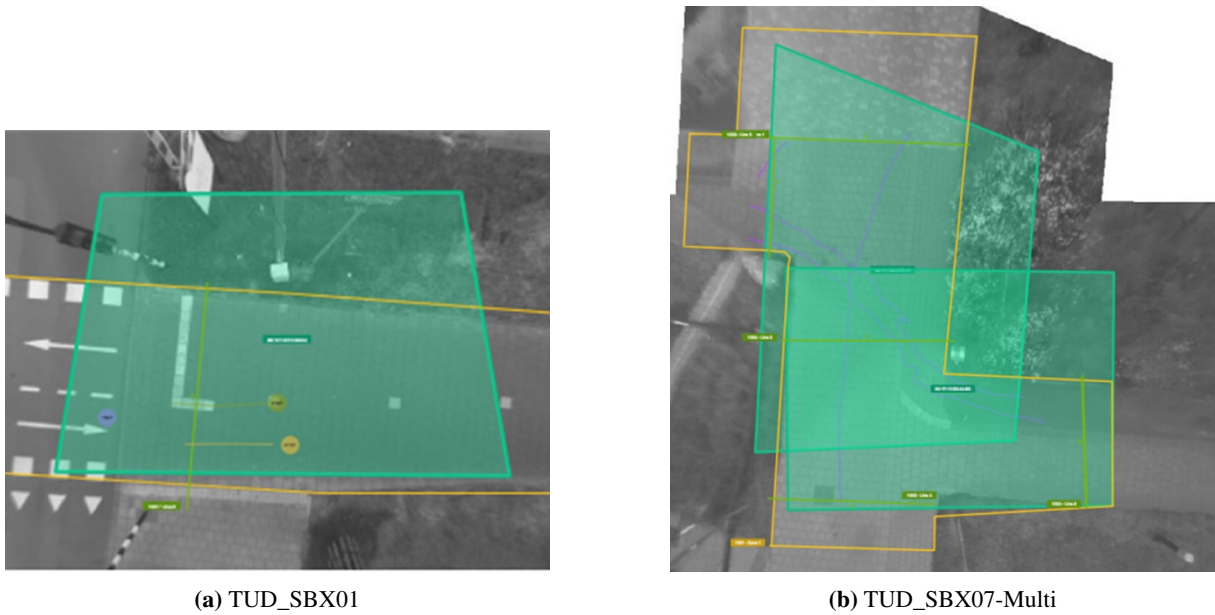


Figure 4.1: Examples of the CREFM camera-based bicycle sensors used for this study. (a) TUD_SBX01, located near the main entrance of the campus next to campus management, and (b) TUD_SBX07-Multi, positioned at CITG. The green lines indicate the virtual detection zones used to count cyclists entering or exiting an intersection. (see Appendix B for larger images)

4.2 Additional features

To aid in the prediction, contextual features are added to enrich the dataset. Extra temporal information is added in the form of bicycle volume this same time last day and last week, this is also known as "lagging". Other contextual data used includes the weather measured from the Davis station on the roof of the EWI building (Schleiss et al., 2025), which provides rainfall, temperature and wind conditions as features. However the dataset states the wind direction is error prone and therefore will not be used in this study. Furthermore features showing exam weeks or vacation weeks and weekdays versus weekend are added to the dataset. These extra features give the predictive models more data to account for variations outside of the infrastructure scope.

These contextual parameters are tested with a grid search, in which each feature can either be included or excluded. The full list of features tested for is:

- temperature
- wind speed
- wind gust speed
- rain amount
- rain intensity
- holiday
- exam week
- weekend indicator

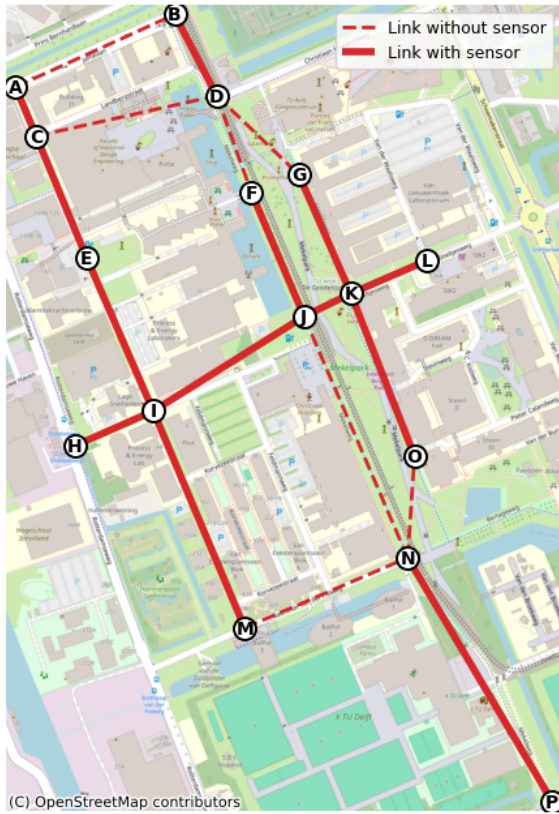
4.3 Train-Test split

To ensure the study is representative predictive models should not be tested on the data they trained on. Because of this the dataset will be split into three subsets: 70% for training, 15% for validation and 15% for testing. This means the parameters within the model are trained with 70% of the data, evaluated and the best hyperparameters found based on the 15% validation data and the final performance evaluated on the totally unseen 15% test data. This crucial step always needs to be done for machine learning models to ensure the generalisability and reliability of the predictions.

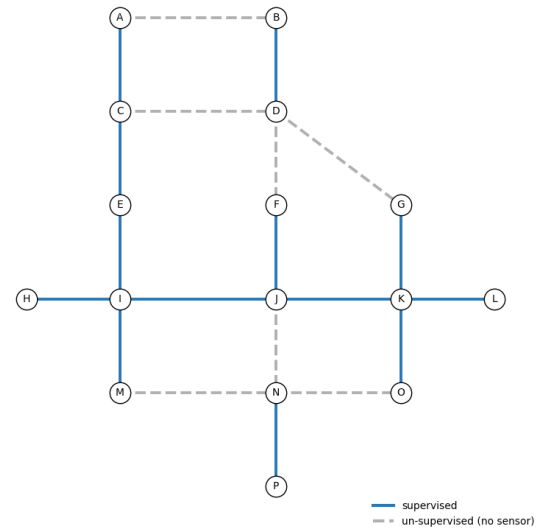
To ensure a balanced representation of weekdays, weekends, holidays and exam periods, the dataset was split on a per-day basis into train, validation and test subsets (70–15–15). Days were randomly assigned while preserving the relative frequency of each calendar category. This is necessary due to holiday and weekends having a big impact on the bicycle counts and the dataset not being big enough to ensure a close representation of the full dataset. Which means, by pure chance only holiday days could be in the validation or test set which would skew results. A fixed random seed was used to guarantee reproducibility.

4.4 Campus Network

To be able to use the GNN model a graph network of the campus is necessary. Using the available sensor locations and the existing intersections on campus a network is made. In Figure 4.2a the nodes and edges of the network are drawn on a map and in Figure 4.2b a schematisation of the network is shown. Because of the limited scope of this study the network will be limited to these nodes, however multiple variations on this base network will be tested. Not every edge has a sensor, edge AB, CD, DG, DF, MN, JN and NO do not have sensors and will therefore be excluded with a mask from the loss function, which means the model will not be penalised for predictions made on these edges.



(a) Geographic network



(b) Schematic used in the model

Figure 4.2: Campus bicycle network representation: (a) map view and (b) schematic. (see Appendix B for enlarged figures)

Chapter 5

Model development

This chapter describes the design and implementation of the predictive models. It explains the structure and function of the historical average, linear regression, and spatio-temporal Graph Neural Network models, as well as the hyperparameter grid search, training procedure, and performance metrics used to optimise them.

5.1 Models

Three different models are used to predict bicycle flow on the campus. As a baseline model a historical average is used that simply returns the average bicycle flow of that time at each time slot. This model serves as a baseline and benchmark in providing a stable and understandable prediction to which other models can be compared. The second model is a linear regression model that incorporates the lagged bicycle counts and contextual factors. This model is easy to interpret and serves as a lightweight comparison against the more complex model.

The third and main model, central to this study, is the spatial-temporal Graph Neural Network (ST-GNN). This model is chosen for its ability to exploit the connectivity of the network while keeping a temporal connection. The model uses graph convolutional layers to capture the spatial structure between intersections, while the temporal dimension is captured by temporal convolutional layers. These layers together allow the neural network to learn how the change of bicycle flow propagates from one node to connected nodes.

Specifically, the model is build up of several layers which are structured as shown in Figure 5.1, the n *times connection is a parameter that can be repeated as many times as deemed fit:

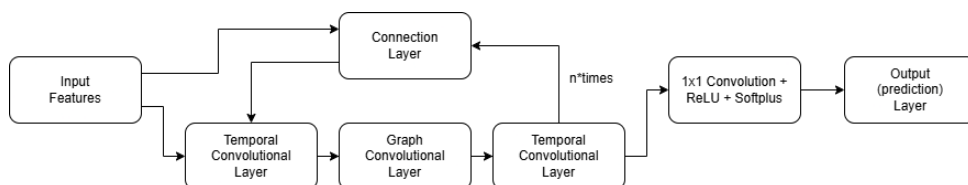


Figure 5.1: Layer architecture of the spatio-temporal GNN combining graph and temporal components. (see Appendix B for larger diagram)

- **Input embedded layer.** The raw features including counts, lagged counts, time schedules and weather conditions are projected into a higher dimensional space to increase the ability of the model to capture complex non linear relationships between the features.
- **Graph convolutional layers (GCN).** These layers provides the neural network with the spatial information and work based on the graph network representation of the campus. It learns the "dependence" between one node and the surrounding nodes. Through more layers, this information

propagates through the network. Enabling the model to capture how bicycle flow at one node impact others.

- **Temporal convolutional Layers (TCNs).** Where the GCN layers are able to learn the spatial interactions, the TCN layers allow the model to learn temporal dependencies. These layers allow the model to learn short and long term temporal patterns, such as drops in bicyclists during rain and reoccurring morning rushes.
- **Connection layers.** These layers connect deeper layers to the input features by bypassing previous layers so information from raw data is not forgotten and used together with learned interactions.
- **Dropout layers.** These layers are integrated within the TCN and GCN layers and are used to mitigate overfitting. These layers randomly set a part of the internal learned parameters to zero forcing partial learning to start again and reduce fitting perfectly to the training data without being able to generalise for test data.
- **Output layer.** This final layer maps the models internal data to a prediction time series for each edge of the network.

5.2 Hyper-parameter grid search

To get the best results, the hyper-parameters should be tuned. Choices such as which optimizer, what learning rate, the number of hidden layers, dropout rates, which loss function to use and to answer the subquestion which contextual parameters should be included need to be tested. A grid search can be done to evaluate all these parameters together to find the best configuration. The grid search is done on the validation set, with reduced amount of epochs to fit to the scope of this study, and includes these tested parameters:

- **Time Granularity:** 5 min, 15 min, 30 min
- **K-hop strength:** 2, 3
- **Hidden Neurons:** 32, 48, 64
- **Number of hidden layers:** 2, 3
- **Dropout rates:** 0.1, 0.2
- **Learning rate:** 10^{-3} , 10^{-4} , 10^{-5}
- **Weight decay:** 0, 10^{-4}
- **Patience:** 8, 12
- **Horizon in:** 12, 24

5.3 Training

In Figure 5.2 we can see the MAE vs Epoch graph. Here we can see the error is large when starting to train and quickly reduces until it approaches a plateau, this shows effective learning. Having a lower training MAE than validation MAE is common and logical, importantly we can see the validation MAE roughly following the training MAE. This shows the model is well generalised. The Training MAE also reaches a plateau which indicates no excessive overfitting is present, this is mostly due to the dropout in the model.

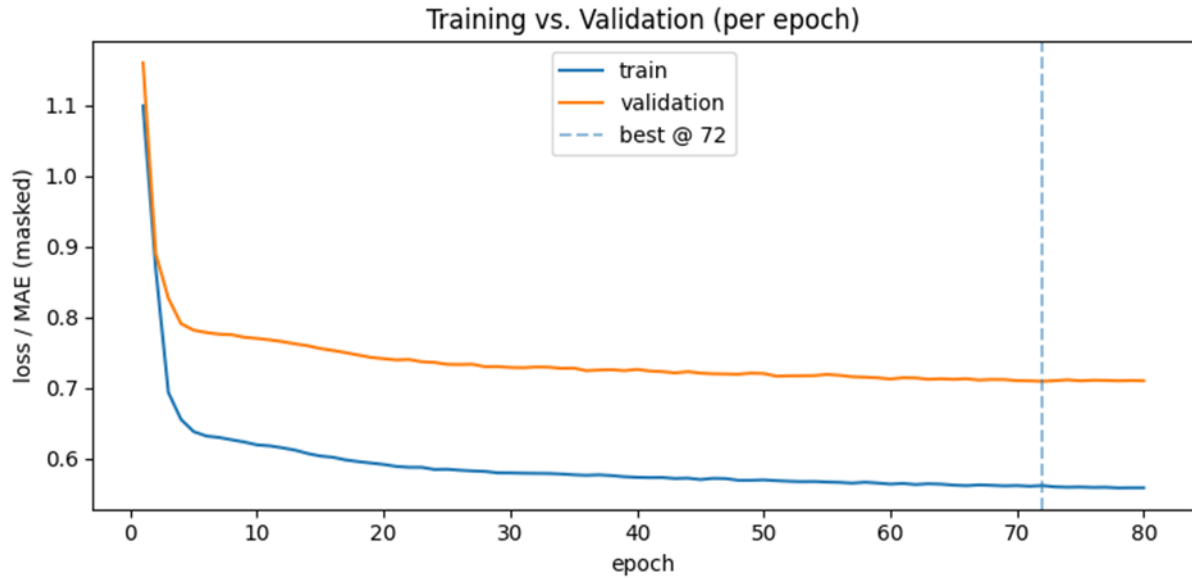


Figure 5.2: Training and validation MAE per epoch. The similar trends indicate good generalisation. (see Appendix B for full-size plot)

Chapter 6

Results and performance evaluation

This chapter presents the results of the model experiments. It shows the predictive accuracy of the different models, the influence of contextual features, and the impact of network configurations. The results are visualised through tables and figures and are compared with findings from relevant literature.

6.1 Network representation

The first subquestion: How should the TU Delft campus bicycle network be represented as a graph for predictive modelling? The base network design was described in Section 4.4 and can be seen in Figure 6.1a. Three other networks derived from the original network are tested as well. The base network achieved a MAE of 0.975. The first derivation, removing node F from the network to eliminate an unsupervised connection (D-F) can be seen in Figure 6.1b with a MAE of 0.972 over the validation set, a marginal improvement over the base network. The third network tested has no A-B connection and no node F, the AB connection is also not supervised. This network can be seen in Figure 6.1c and achieved a MAE of 0.997, which is worse than the previous two. The fourth and last tested network incorporates the removal of F, no A-B connection and removes the unsupervised N-O connection as can be seen in Figure 6.1d, this model only achieved a MAE of 1.002. These results show that network two, without the F node results in the lowest MAE on the validation set.

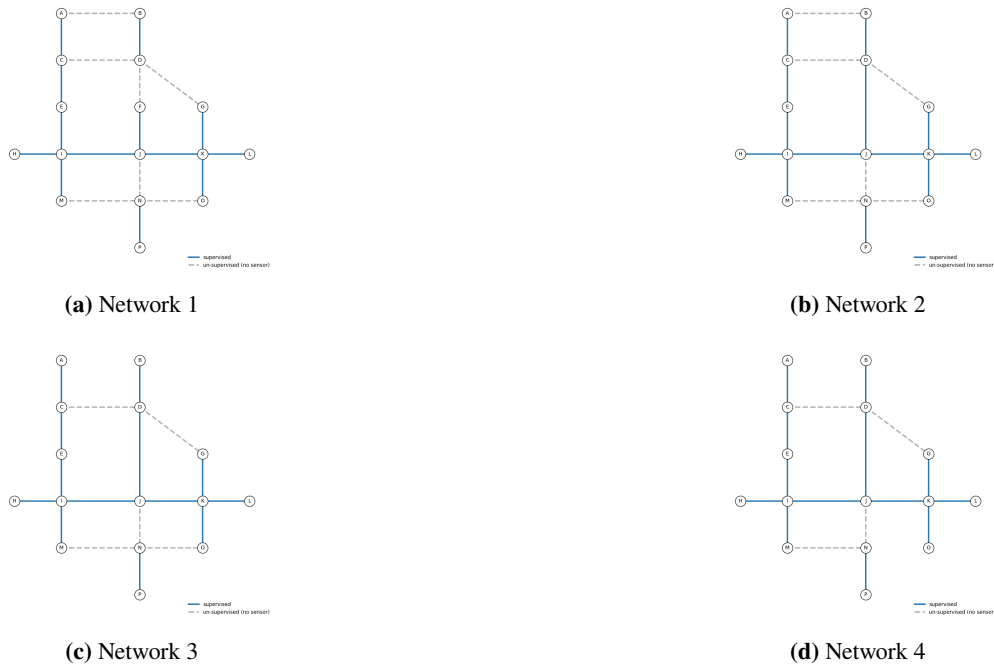


Figure 6.1: Tested network configurations (1–4) used in the experiments. (see Appendix B for larger maps)

6.2 Role of contextual features

Subquestion two, What use of contextual factors such as rain or exam weeks can improve GNN predictions? To answer this subquestion the value of each contextual feature is measured by training with and without each feature. In Table 6.1 we can see the average MAE with the corresponding feature included.

Table 6.1: Validation MAE of the GNN for each contextual feature, compared with baseline validation MAE.

Feature	Only Feature MAE	Avg. MAE When Included
temperature	0.9221	0.9573
wind gust speed	0.9575	0.9604
holiday	0.9645	0.9632
wind speed	0.9913	0.9639
examday	0.9431	0.9641
weekend	0.9517	0.9646
rain amount	0.9326	0.9651
rain intensity	1.0305	0.9655
(none)	0.9720	0.9720

Table 6.1 shows the contextual features rain intensity and wind speed do not improve predictions significantly. The most impactful features for GNN predictions are whether it is exam week and the temperature. The combination with the best validation MAE out of the 256 combinations tested is:

- Temperature
- Rain intensity
- Wind gust speed
- Weekend indicator
- Exam week

This combination achieved a MAE of 0.8997 on the validation set. This being the best combination answers the sub-question two and this combination will be used to evaluate the final model after a grid search of hyperparameters.

6.3 Model comparison

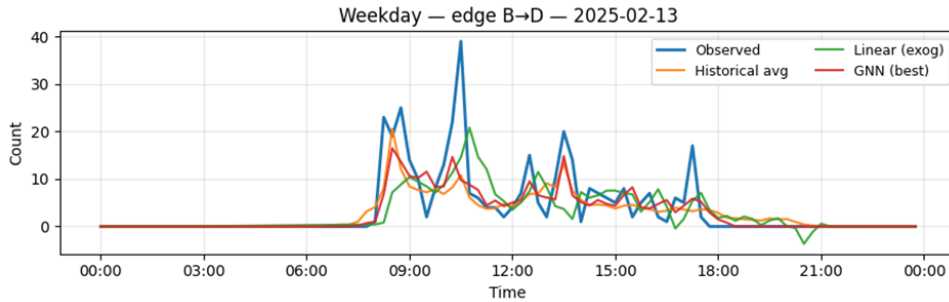
The Three models, historical average as baseline, linear regression with contextual features incorporated and the GNN with temporal and spatial structure learning. The three models are tested on the unseen test set and show the MAE and RMSE respectively in Table 6.2. We can also see the percentage error change compared to the baseline. It shows an already low error historical average model, which is improved on marginally with the linear regressor. The GNN model shows a significant improvement over the baseline model.

Table 6.2: Model performance on the test set. Lower is better. Baseline is the Historical average (100%).

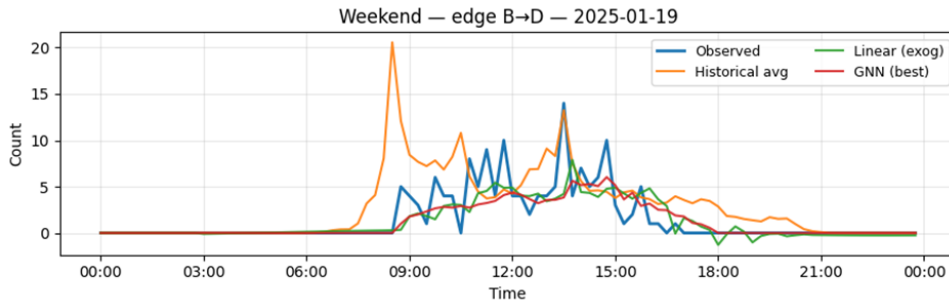
Model	MAE	RMSE	MAE (% of baseline)
Historical Average	0.900	2.220	100.0%
Linear Regression	0.851	2.027	94.5%
ST-GNN	0.629	1.682	69.9%

Picking a random week and weekend day and plotting all the predictions we can see the three models versus the observed ground truth in Figures 6.2a and 6.2b. Figure 6.2a shows that on the weekday, all three models follow the characteristic TU Delft campus rhythm at least moderately, with sharp morning and midday peaks linked to lecture start times. The historical average baseline (orange) follows these peaks but smooths them out more, resulting in often lower than observed peaks. The linear regressor (green) captures the peaks and valleys but often misaligns them in time. The GNN model (red) provides the best overall fit, tracking the gradual rise and fall of bicycle counts throughout the day and remaining close to the observed curve except for the steepest spikes.

In Figure 6.2b on the weekend, bicycle activity is lower and less structured. Here, the historical average perform very poor because it overestimates traffic based on weekday patterns. Both the linear and GNN models adjust to the reduced flow, but the GNN adapts more effectively, maintaining a realistic shape that mirrors observed weekend variability. This contrast highlights the GNN’s strength in generalising across different temporal contexts, while simpler models rely too heavily on periodic weekday patterns.



(a) Weekday bicycle counts: observed vs. predicted.



(b) Weekend bicycle counts: observed vs. predicted.

Figure 6.2: Observed and predicted bicycle counts for typical weekday and weekend conditions. The weekday graph shows clear morning and midday peaks aligned with lecture start times, which the ST-GNN follows closely. During weekends, overall activity decreases and becomes more irregular; the ST-GNN adapts to these lower volumes more effectively than the simpler models. (see Appendix B for full-size plots)

6.4 Comparison with literature

Previous studies have shown that hybrid GNN's can gain error improvements over baseline, Yu et al. found their ST-GNN model scored a MAE of 2.25 against a historical Average of 4.01, a 43.9% error reduction (Yu et al., 2018). Li et al. show their comparable DCRNN model scored a MAE of 2.77 against the historical average of 4.16, which is a reduction of 33.4% (Li et al., 2018). The results from this study show a MAE of 0.629 on the ST-GNN model and a MAE of 0.900 on the historical Average baseline, this gives a reduction of 30.1% in MAE.

6.5 Visualisation of the results

Different ways to visualise a traffic network are possible, in Figures 6.2a and 6.2b of Section 6.3 a time series shown on where the predictions over one direction on one edge is shown. This visualises the connection between bicycle count and time dependencies. However the model predicts a whole network, to show the spacial dependencies a network graph as shown in ?? can be used. The top network in the figure shows the observed bicycle count for every supervised edge at the given time, in this case on 11-02-2025 at 8:45 in buckets of 15 minutes (8:45-9:00). Below that the predicted count by the ST-GNN model is shown for that moment and the remaining network shows the error between the two. It shows that the largest error occurs from B to D as these are bycyclist that are not seen in the network before and therefore not anticipated by the spatial part of the ST-GNN.

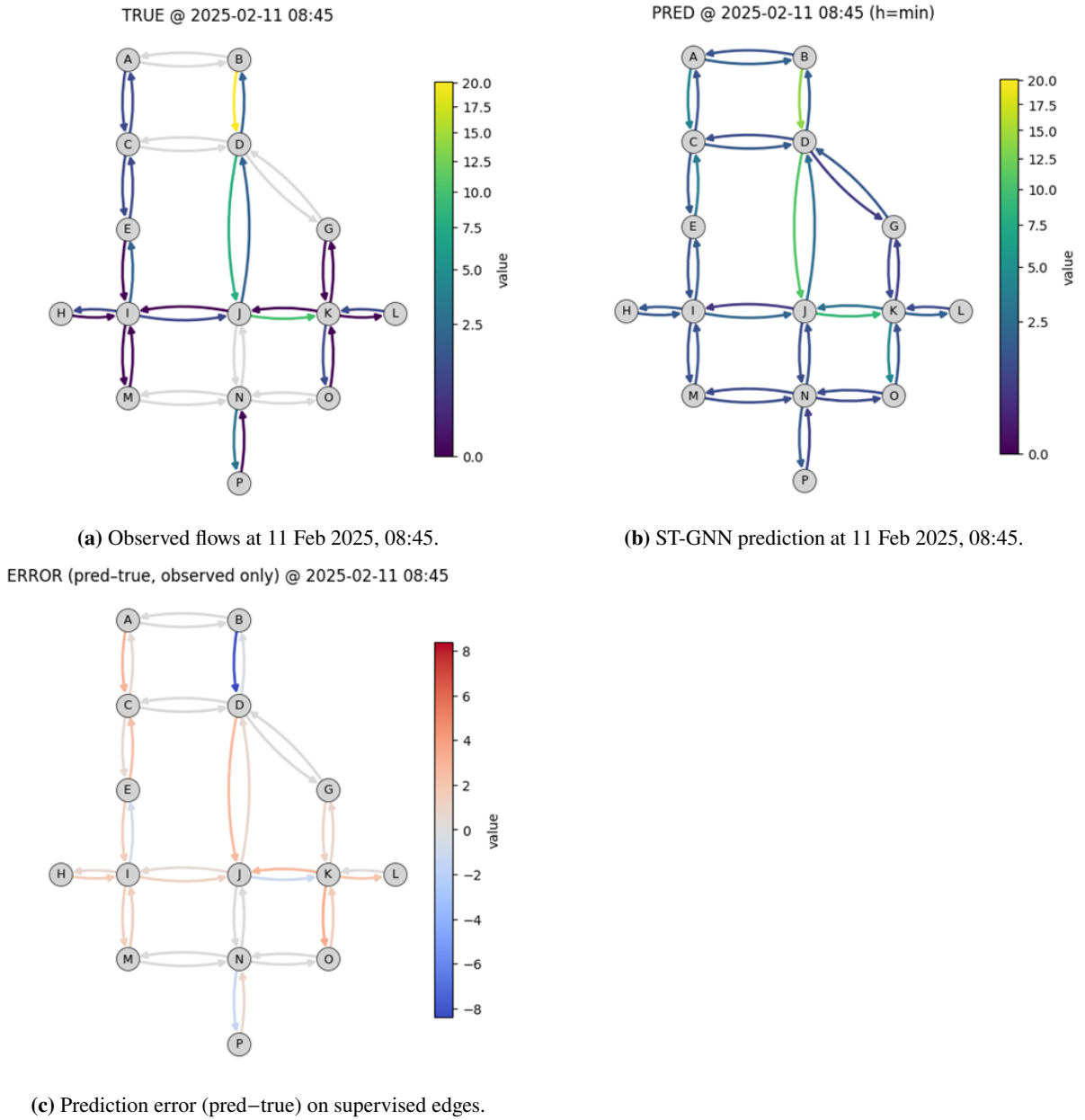


Figure 6.3: Network flows at 11 February 2025, 08:45. (a) observed counts; (b) ST-GNN prediction; (c) error (prediction minus observation). See Appendix B for full-size panels.

Chapter 7

Discussion

This chapter interprets the results and critically reflects on the study's assumptions and limitations. It discusses the implications for bicycle traffic forecasting and evaluates the reliability and generalisability of the conclusions. Recommendations for future methodological improvements are also outlined.

7.1 interpretation of the results

The results show multiple things, firstly Section 6.1 shows even small changes to the network have impact on the prediction capability of the ST-GNN model, which suggests the spacial dependencies are important to model the bicycle flow more accurately. This can also be seen in Section 6.5, where it shows the largest error rates are at edges where new bicyclists enter the network and the count therefore can not use the spacial dependency yet and has to rely on the temporal layers. These findings together with ST-GNN's improved performance over the historical average and linear regressor models it shows the improved value of the spacial dependencies this model provides.

The grid search over the contextual features shows the impact of adding contextual features are significant. Temperature and wind gust speed are the features with the most consistent improvements. rain amount and intensity do not improve performance as much. The holiday feature not performing as expected could be caused by the holiday days in the validation set also being weekend days. The bicycle flow counts between weekend days and holidays does not differ much and the model might have preferred the weekend feature over the holiday feature as a result. The final best tested combination is temperature, rain intensity, wind gust speed and whether it is a weekend and exam day. This combination's MAE on the validation set of 0.8997 is a significant improvement over the MAE of 0.972 without contextual features.

The ST-GNN model shows a good improvement over the linear regression model and the historical average baseline. This means the added complexity of taking temporal and spatial dependencies into account does improve the prediction significantly. The improvement of 30.1% in MAE is comparable with the error rate of related GNN studies done on different traffic forms. With this study having a comparatively short time and low resources this error rate could very well be improved on significantly and invoke the thought if ST-GNN models with contextual features are more suited for bicycle traffic flow predictions than other forms.

7.2 limitations/assumptions

The conclusions of this research should be interpreted with caution. The dataset covers only a fraction of the academic year and may therefore not contain enough variation to generalise beyond winter and early spring conditions. As a result, the model has not been exposed to the full range of weather and behavioural variability that typically characterises cycling in the Netherlands. The limited number of

sensors further constrains spatial learning. While the GNN can propagate information between measured edges, large unsensed gaps in the network reduce it's ability to model the complete flow from start to end. The masked loss function helps prevent overfitting to this missing data but simultaneously reduces the ability of the model to learn indirect relationships through unsupervised links.

At the methodological level, several assumptions also shape the results. The input data were globally normalised rather than scaled per sensor. This amplified magnitude differences between high-volume and low-volume intersections and likely biased training towards the busiest edges. A per-sensor normalisation strategy would reduce this imbalance and allow the model to learn relative rather than absolute flow patterns.

The mean absolute error (MAE) loss was weighted by activity to avoid trivial zero predictions while maintaining interpretability and robustness. Nevertheless, because MAE assigns equal importance to all deviations, it may still under-represent sharp peak flows that are most critical for capacity planning. Hybrid losses such as the Huber loss could help strike a balance between stability and peak sensitivity.

The set of contextual features used such as temperature, wind, rain and calendar features improved results but could be extended further. Including additional environmental and behavioural variables could further improve performance. Moreover, the feature set was primarily optimised for the GNN, not for the linear regression baseline, which may have contributed to the performance gap between these models.

The grid search on the features and parameter optimisation were performed after each other, this assumes the optimal contextual features are independent from the model parameters. However this interaction could be significant, and other features would be selected if done after the parameter search. A combination of parameter to feature to parameter grid search could be better and a full grid search of both features and parameters combined is the most accurate way, but due to exponential computational cost would approximately take 130 days, longer than is allocated for this whole study.

This limitation directly ties into the main restriction of the study, runtime and computational limitations constrained the model complexity and experimentation. The models were trained on a single workstation, with a GTX 1080Ti as GPU, which limited the feasible network size, parameter tuning, and number of validation runs. With more powerful hardware or access to GPU resources, larger architectures and longer time horizons could be explored, potentially improving both performance and stability.

Chapter 8

Conclusions and recommendations

This chapter synthesises the main findings of the study and answers the central research question: to what extent bicycle traffic flow on the TU Delft campus can be predicted and what added value Graph Neural Networks provide over other approaches. It summarises the study's contributions, reflects on its practical and scientific significance, and provides recommendations for future research and implementation.

8.1 Conclusion

This study set out to answer: **To what extent can bicycle traffic flow on the TU Delft campus be predicted, and what value can Graph Neural Networks provide over other approaches?** The motivation behind this came from the need for better bicycle count predictions for use in infrastructure, maintenance and capacity management. The answer was achieved by developing and comparing three predictive models, a historical average, a linear regressor and a ST-GNN model.

The Results show that bicycle traffic can indeed be predicted with good accuracy measured in MAE compared to the baseline historical average, a 28% decrease. This decrease corresponds with similar studies done on car traffic (Yu et al., 2018; Li et al., 2018; Lu et al., 2020) and shows its validity as a effective model. The graphs show good generalisation but poor fitting to schedule induced peaks for the ST-GNN model. Contextual factors did play a large role in improving the model, with the most impactful combination being temperature, wind, rain and whether it is weekend or not.

So yes, bicycle traffic flow on the TU delft campus can be predicted to a comparable extent as other studies with car traffic with the use of GNN's. Also this study shows the GNN model provides a 30.1% and 26.1% error value decrease over historical average and linear regression approaches respectively. It achieves this by taking the spacial and temporal dependencies directly into account and combining them with the contextual features.

8.2 Recommendations

1. Extend the dataset and improve sensor coverage. The most immediate improvement lies in expanding both the temporal and spatial extent of the dataset. The present data only covers four months and a limited number of sensor locations, which restricts the model's ability to capture seasonal trends and long-term behaviour. Future datasets should span at least a full year to include winter and summer cycling patterns, as well as transitional months where weather effects are most visible. Additional sensors should be installed at critical intersections currently unsupervised, particularly near the Mekelpark, the South Campus access routes, and the northern entrance near the faculties of TPM and Architecture, where bicycle flow converges or diverges sharply. A denser network will allow the GNN to better use spatial connections and reduce information loss due to masked edges.

2. Use larger and more expressive models. This study was limited by computational constraints, running models locally on a single workstation. With access to more computational resources, deeper and wider architectures could be explored. Larger models, or more computationally heavy layers such as GRU or LSTM for the temporal layers, could improve the ability to represent complex temporal interactions and capture sharper peaks. Training on multiple GPUs or cloud resources would make it feasible to experiment with larger models, longer time horizons, and alternative loss functions such as the Huber or quantile loss, which may better handle extreme values.

3. Compare with locations less tied to schedule-driven mobility. The TU Delft campus is a unique environment where bicycle traffic is heavily influenced by the university schedule. This makes it a useful but also atypical test case. To assess generalisability, the model should be tested on urban networks where cycling demand depends on more general traffic patterns rather than fixed lecture times. Applying the same GNN framework to the city centre in Delft, The Hague or Utrecht would reveal whether the method performs equally well when temporal patterns are weaker and variability higher.

Chapter A

Word explanations

This appendix provides concise definitions of key technical terms, abbreviations, and concepts used throughout the thesis. It is intended to support readers who are less familiar with specific terminology from data science or transport modelling.

Graph Neural Network (GNN) A type of neural network that operates directly on graph-structured data. It learns how information propagates between nodes (intersections) and edges (connections) in a network.

Spatio-temporal GNN (ST-GNN) A GNN that also incorporates temporal layers to model how traffic patterns evolve over time in addition to spatial dependencies between network elements.

Node A point in the network graph representing an intersection or sensor location.

Edge A link between two nodes representing a bicycle path or connection on the campus network.

Temporal features Variables describing time-related aspects of the data, such as time of day, day of week, or season.

Contextual features External factors that influence bicycle traffic, such as weather, holidays, or exam weeks.

Lagged features Historical observations (e.g. bicycle counts one day or one week earlier) used as input to improve short-term prediction accuracy.

Train-validation-test split The division of a dataset into subsets for model training, hyperparameter tuning, and independent evaluation to ensure generalisability.

Mean Absolute Error (MAE) An evaluation metric measuring the average magnitude of prediction errors, regardless of direction.

Root Mean Squared Error (RMSE) A metric similar to MAE but more sensitive to large errors, as deviations are squared before averaging.

Overfitting A situation in which a model performs well on training data but poorly on unseen data because it has learned noise rather than general patterns.

Dropout A regularisation technique in neural networks that randomly disables part of the model's neurons during training to prevent overfitting.

Hyperparameter A model configuration value (e.g. learning rate, number of layers) chosen before training to influence model performance.

Grid search A systematic method for testing multiple hyperparameter combinations to identify the best-performing model configuration.

Normalization The process of scaling data to a comparable range (for example, between 0 and 1) to improve model stability and convergence.

Masking Excluding specific data points or connections (such as unsupervised edges) from the model's loss function to avoid penalising predictions without ground-truth data.

Loss function The mathematical expression that measures the difference between the model's predictions and the actual observed values during training.

Epoch One full pass through the entire training dataset during the learning process.

Activation function A mathematical function applied to neurons within a neural network that introduces non-linearity and helps the model learn complex relationships.

Reproducibility The ability for another researcher to replicate the same results using the same data and methodology.

Chapter B

Extended Figures

This appendix contains full-size versions of the key figures presented in the main report. They are included here to allow detailed examination of network layouts, visual results, and model performance outputs that were scaled down in the main text for readability.

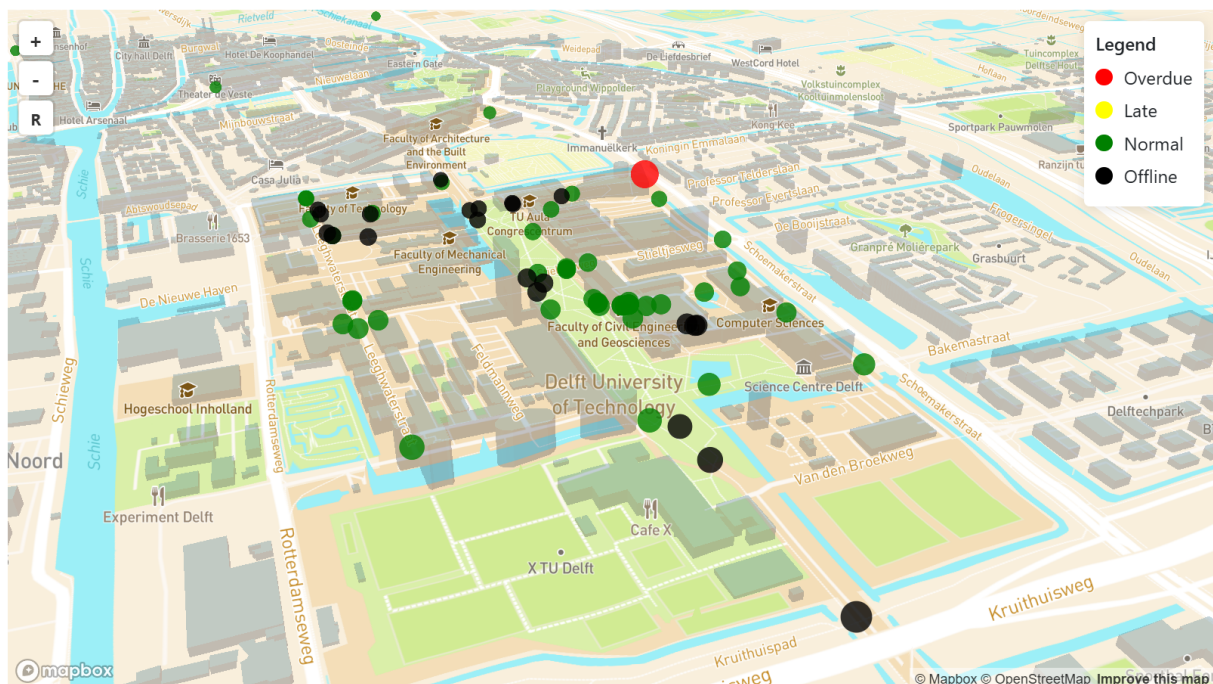


Figure B.1: Full-size version of Figure ??: spatial distribution of CREFM bicycle sensors across the TU Delft campus.

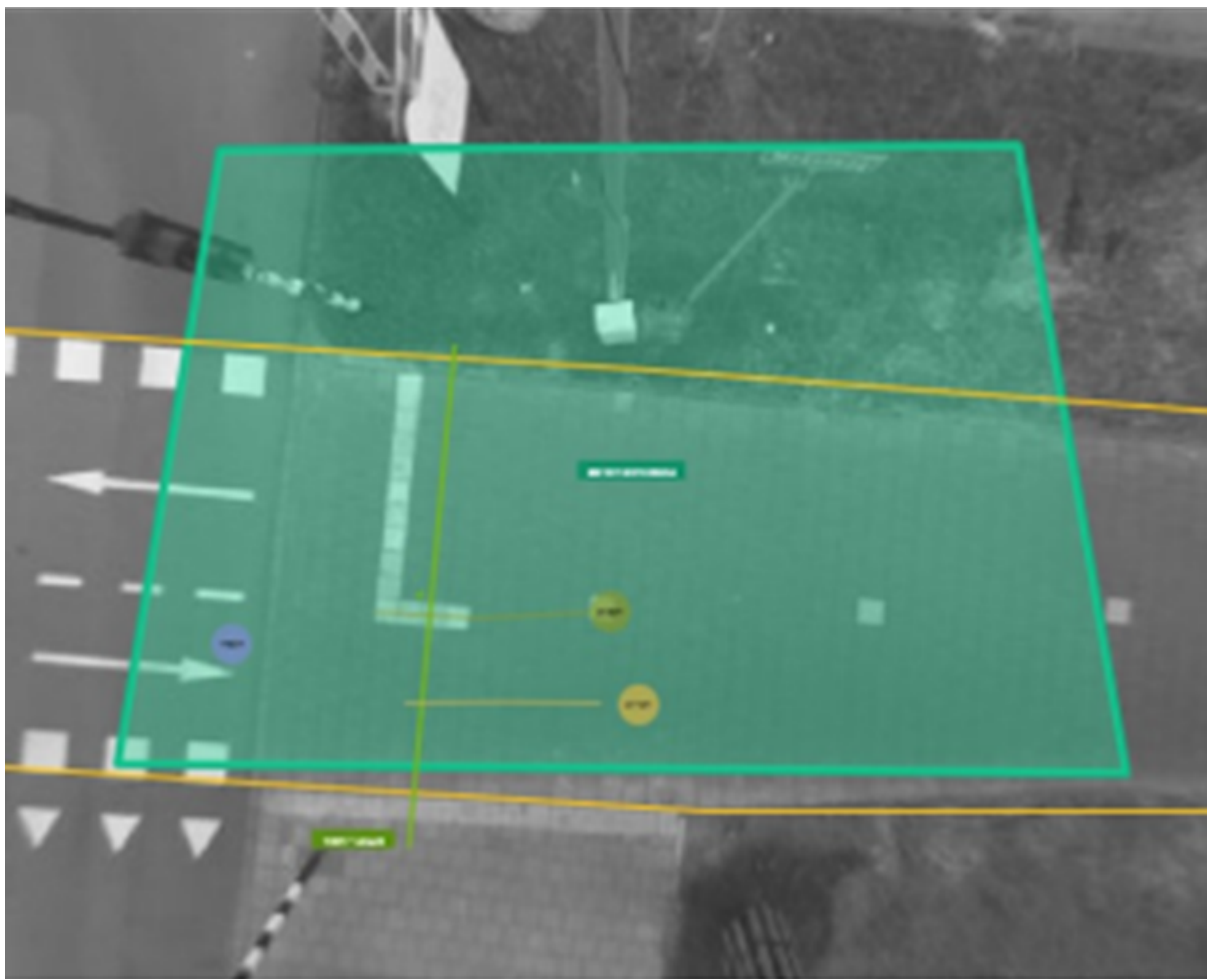


Figure B.2: Full-size version of Figure 4.1: CREFM sensor TUD_SBX01 showing the virtual detection line used to count bicycles.

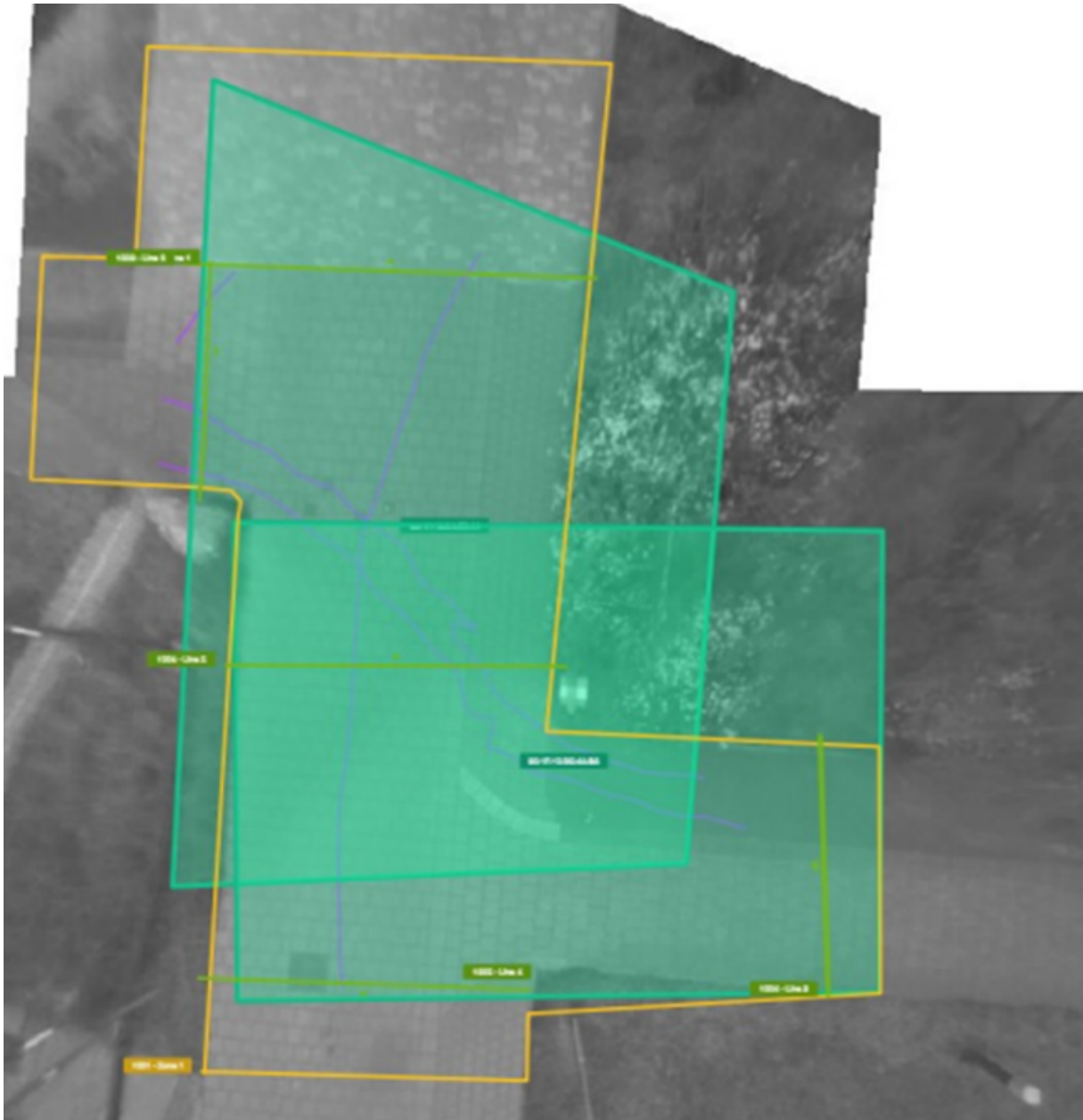


Figure B.3: Full-size version of Figure 4.1: CREFM sensor TUD_SBX07-Multi positioned along the southern campus route.

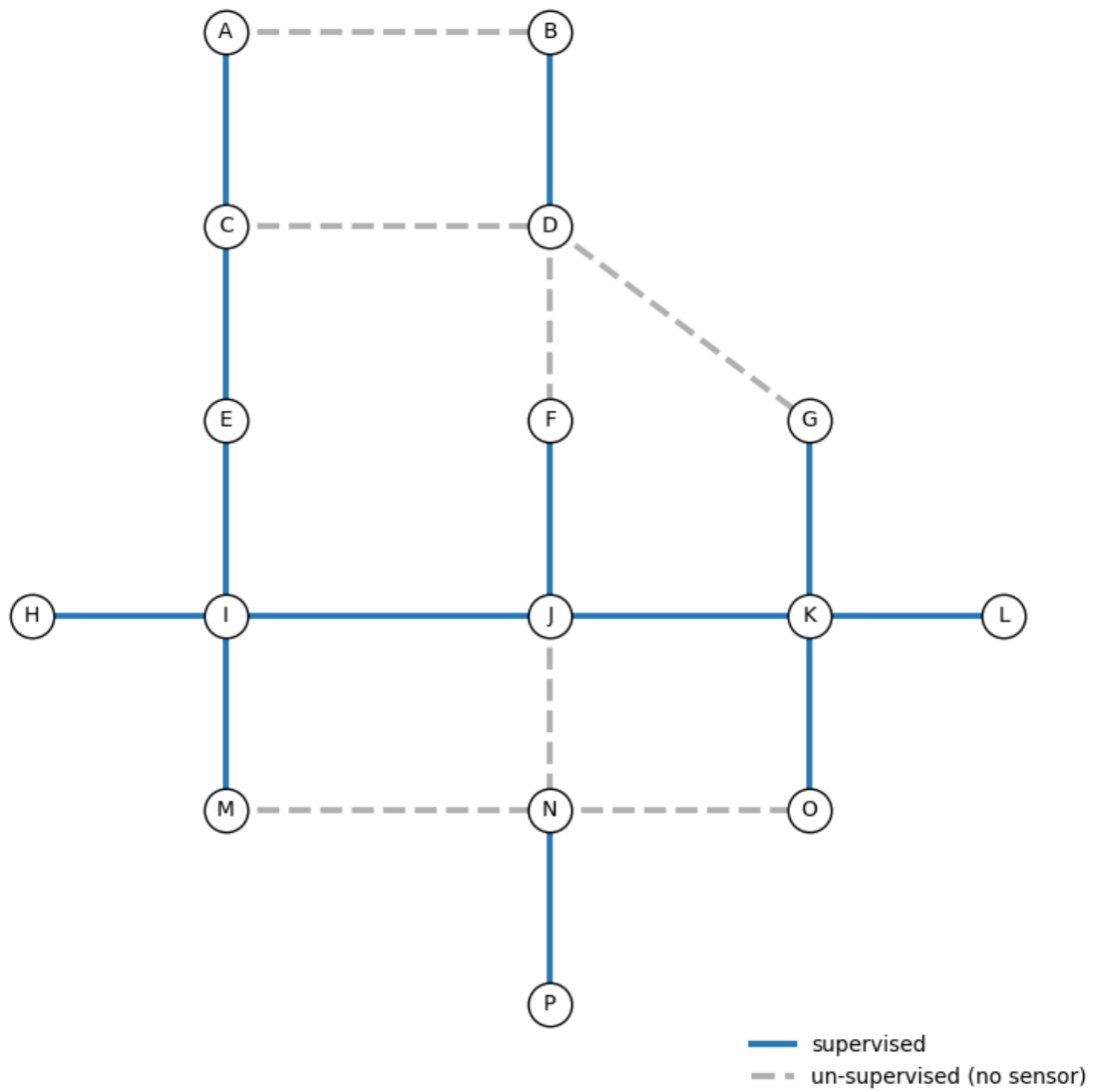


Figure B.4: Full-size version of Figure 4.2: geographic representation of the TU Delft bicycle network 1 including all measured intersections and paths.

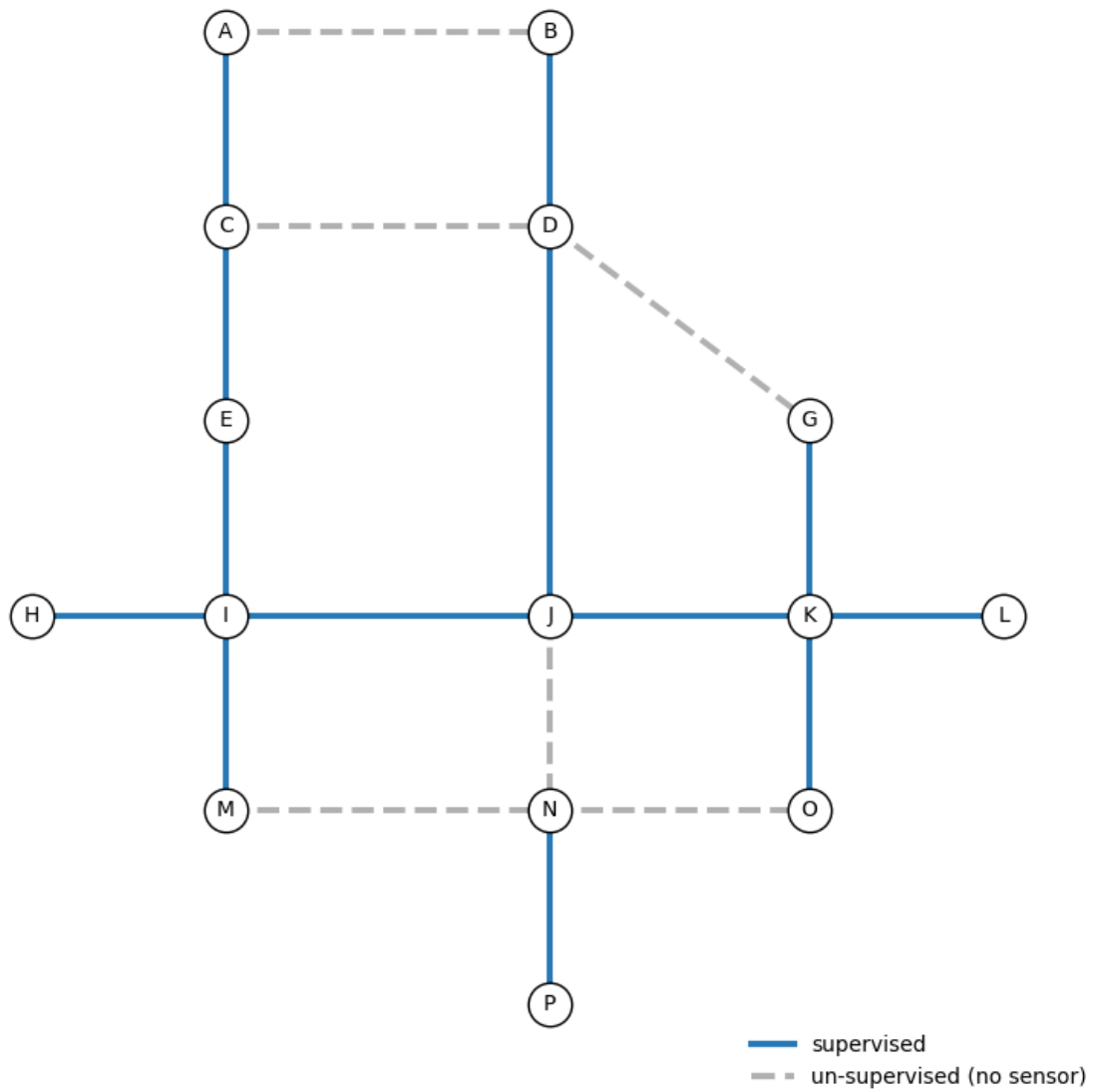


Figure B.5: Full-size version of Figure 4.2: geographic representation of the TU Delft bicycle network 2 including all measured intersections and paths.

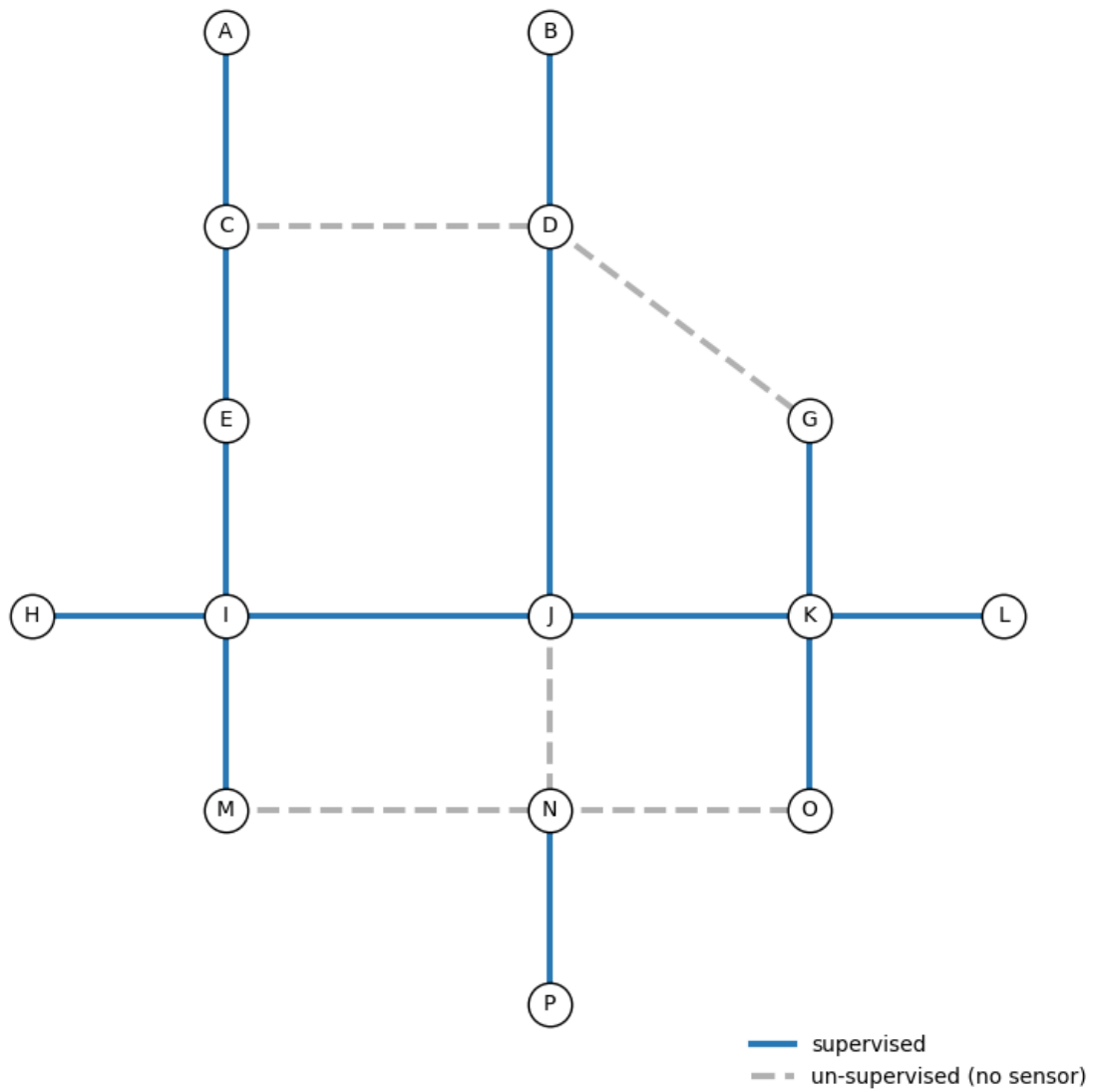


Figure B.6: Full-size version of Figure 4.2: geographic representation of the TU Delft bicycle network 3 including all measured intersections and paths.

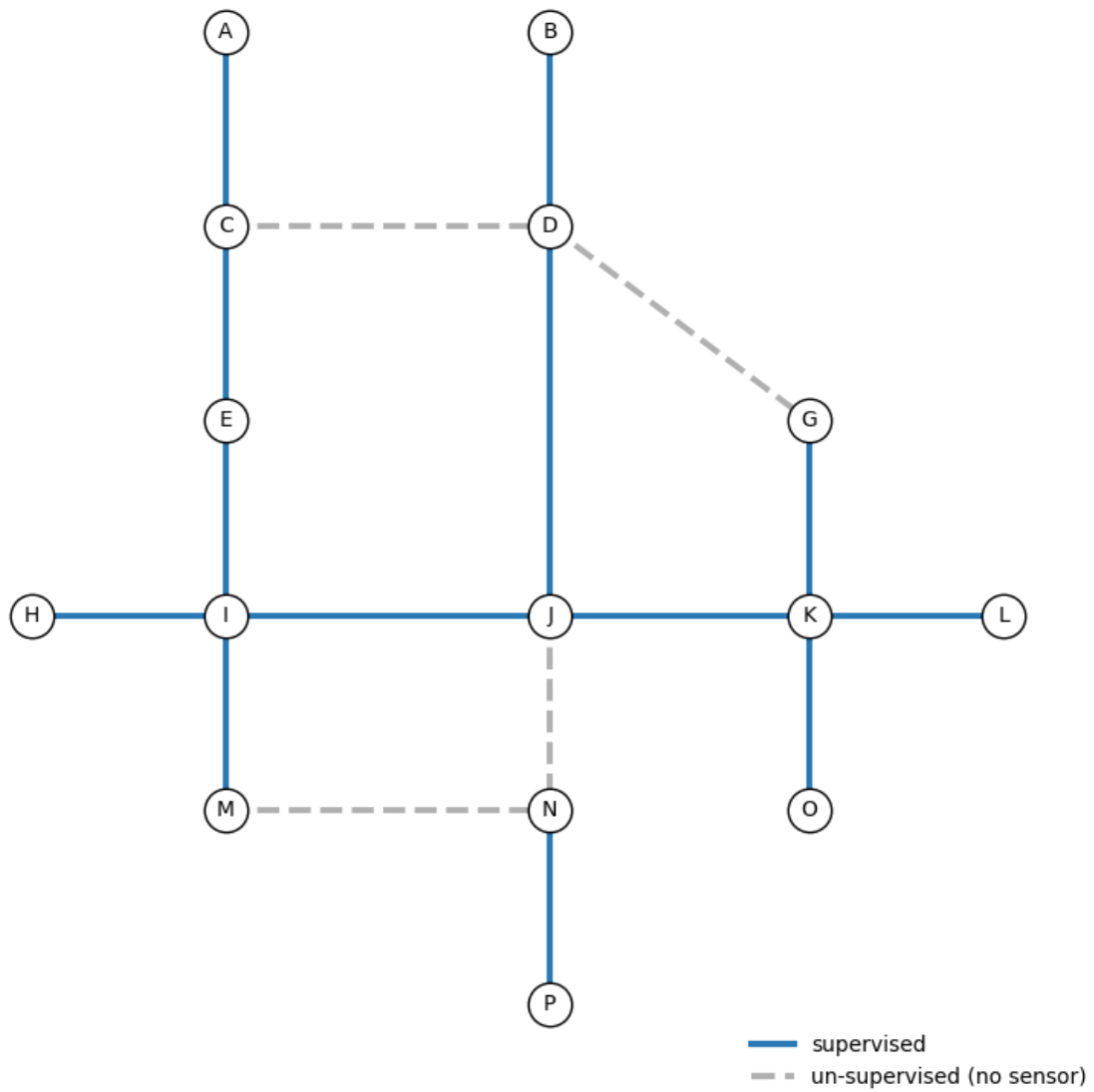


Figure B.7: Full-size version of Figure 4.2: geographic representation of the TU Delft bicycle network 4 including all measured intersections and paths.

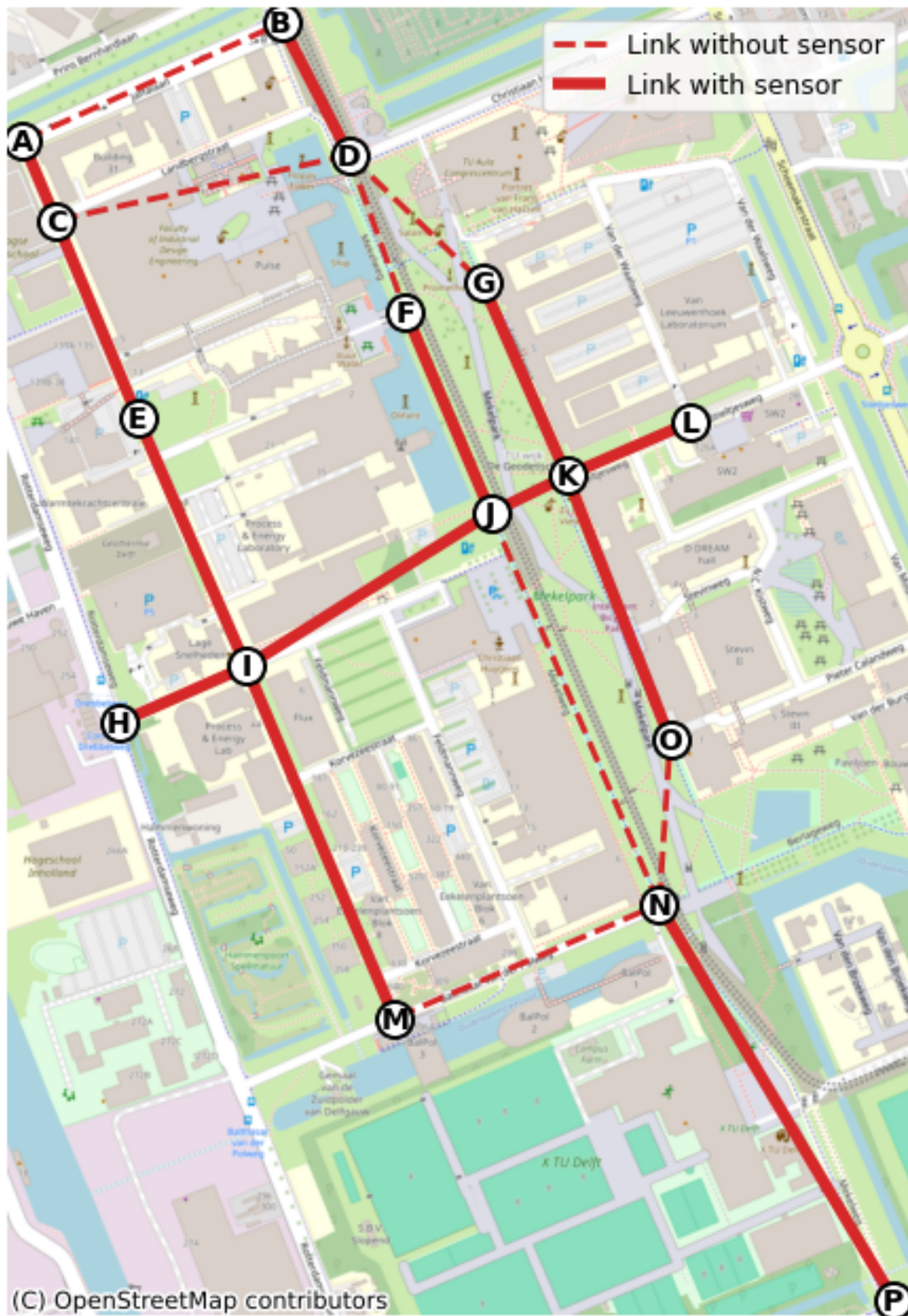


Figure B.8: Full-size version of Figure 4.2: schematic representation of the TU Delft campus bicycle network used for graph modelling.

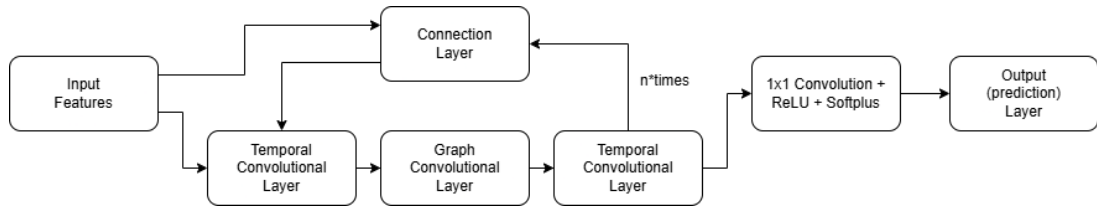


Figure B.9: Full-size version of Figure 5.1: structure of the spatio-temporal Graph Neural Network showing the flow of information between graph and temporal layers.

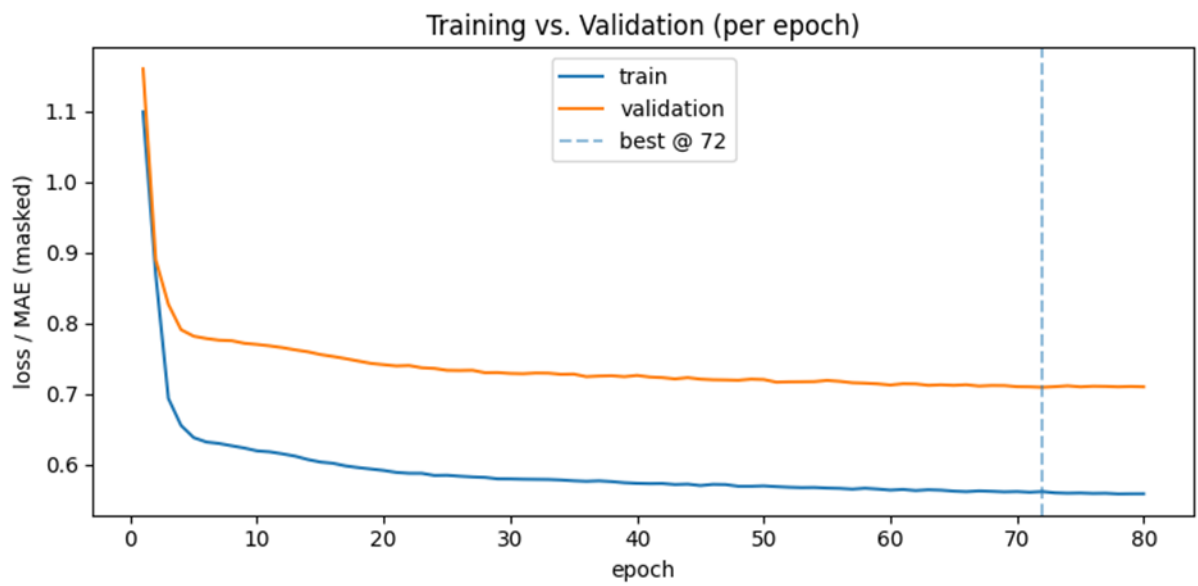


Figure B.10: Full-size version of Figure 5.2: model training and validation MAE per epoch.

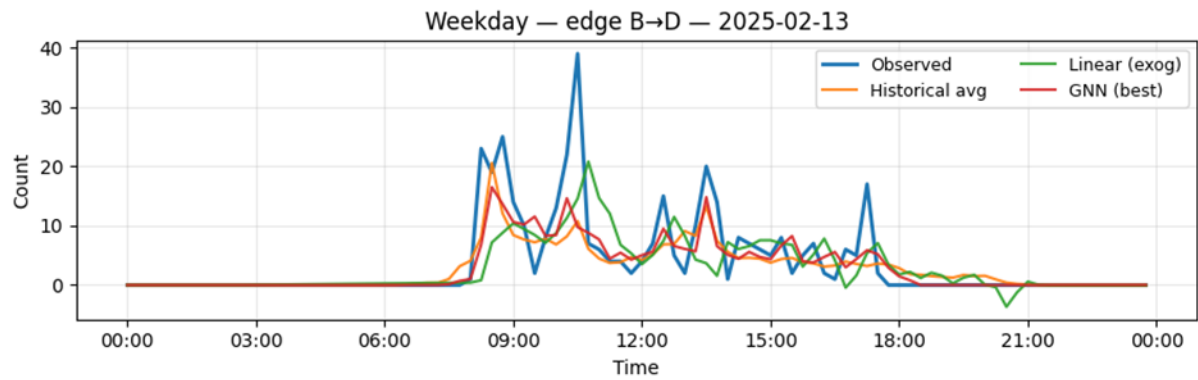


Figure B.11: Full-size version of Figure 6.2a: comparison of observed and predicted weekday bicycle flows for all models.

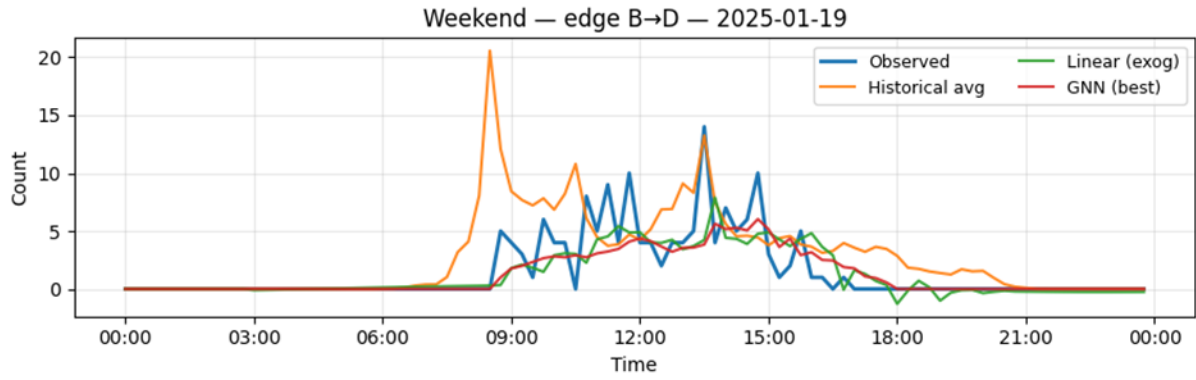


Figure B.12: Full-size version of Figure 6.2b: comparison of observed and predicted weekend bicycle flows for all models.

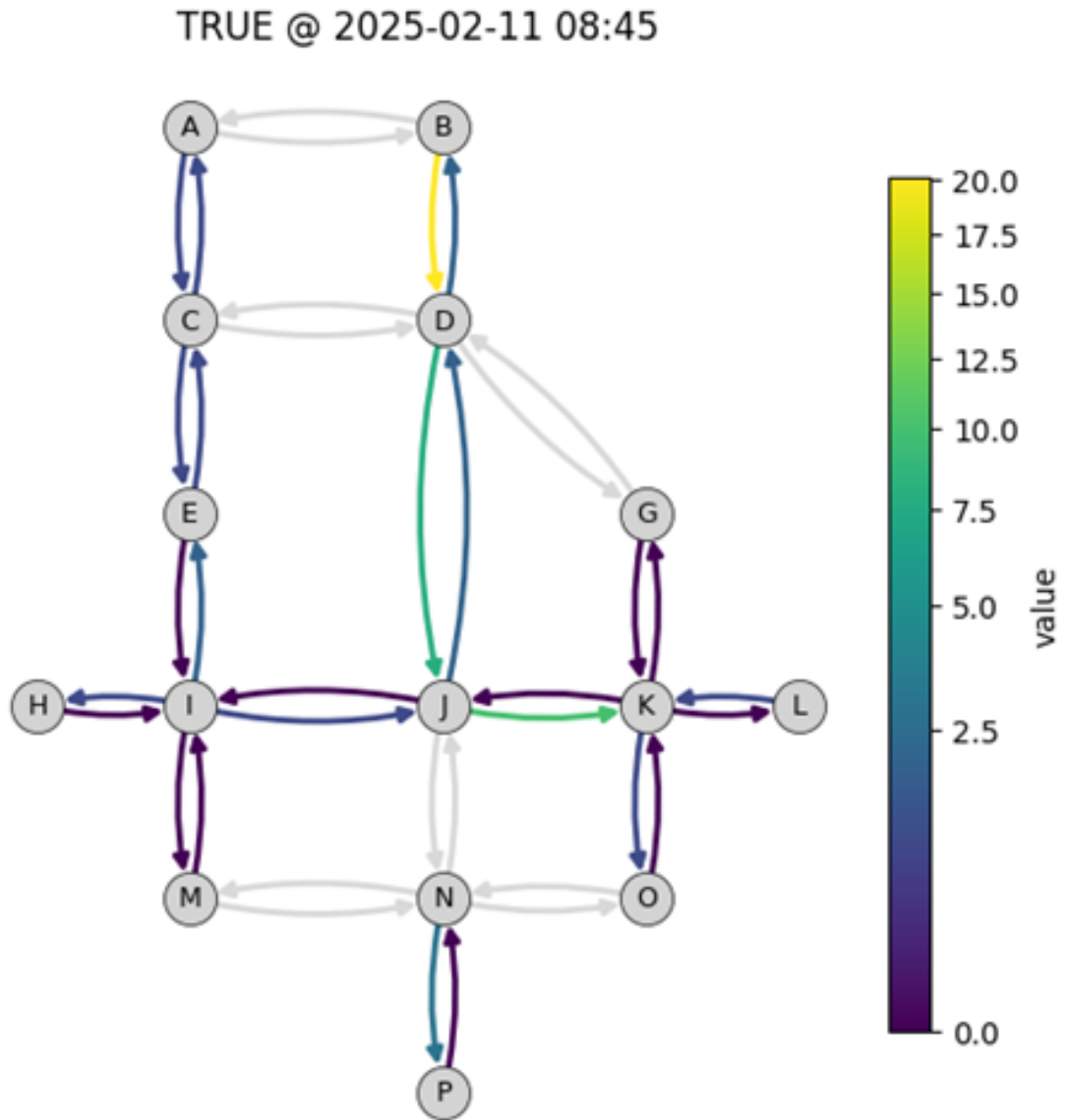


Figure B.13: Full-size version of Figure 6.3a: observed bicycle flows on the TU Delft network at 11 February 2025, 08:45. Edge colours indicate measured counts on supervised links; unsupervised links are shown in light grey.

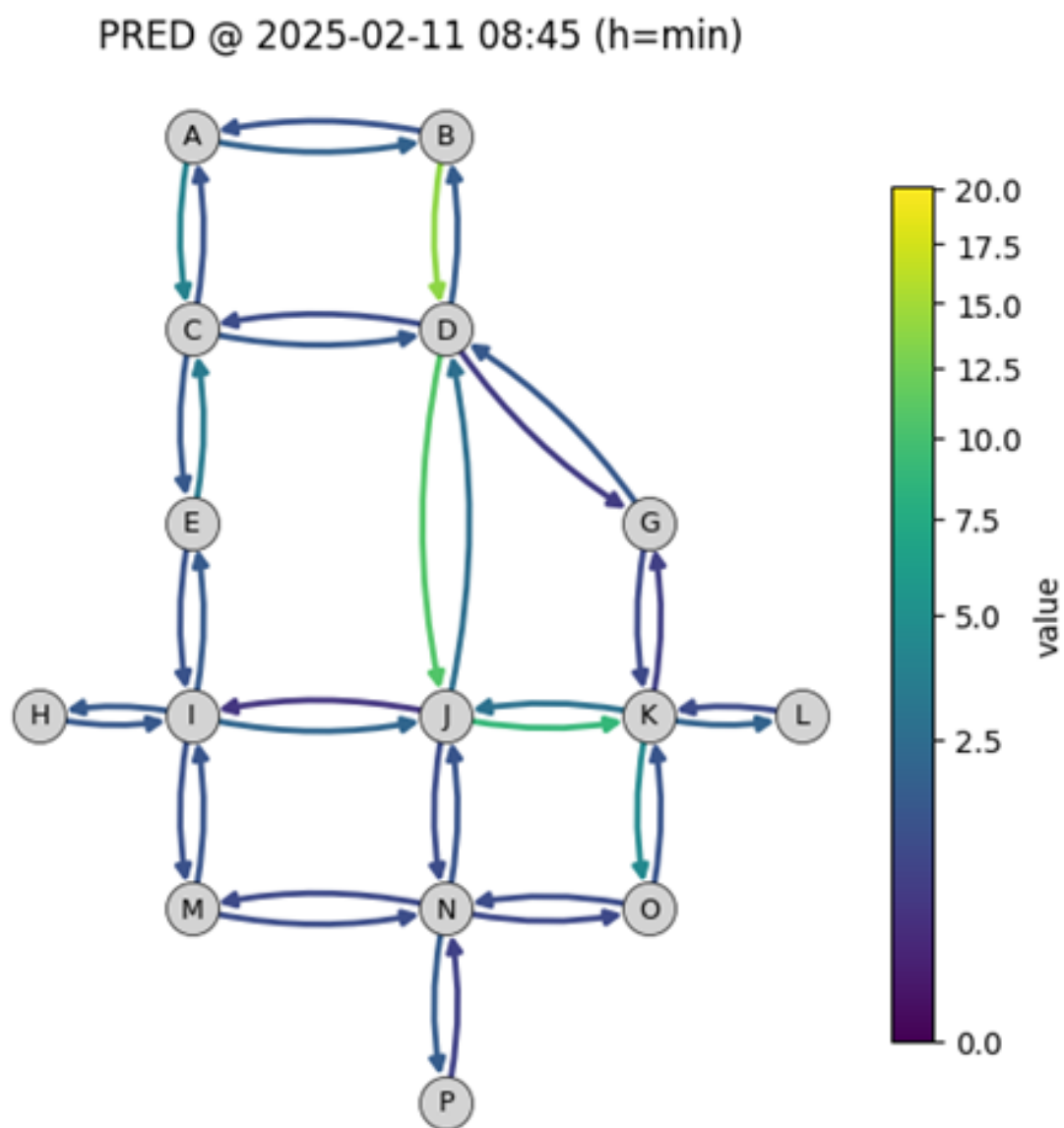


Figure B.14: Full-size version of Figure 6.3b: ST-GNN predicted bicycle flows at 11 February 2025, 08:45 (same colour scale as observations).

ERROR (pred-true, observed only) @ 2025-02-11 08:45

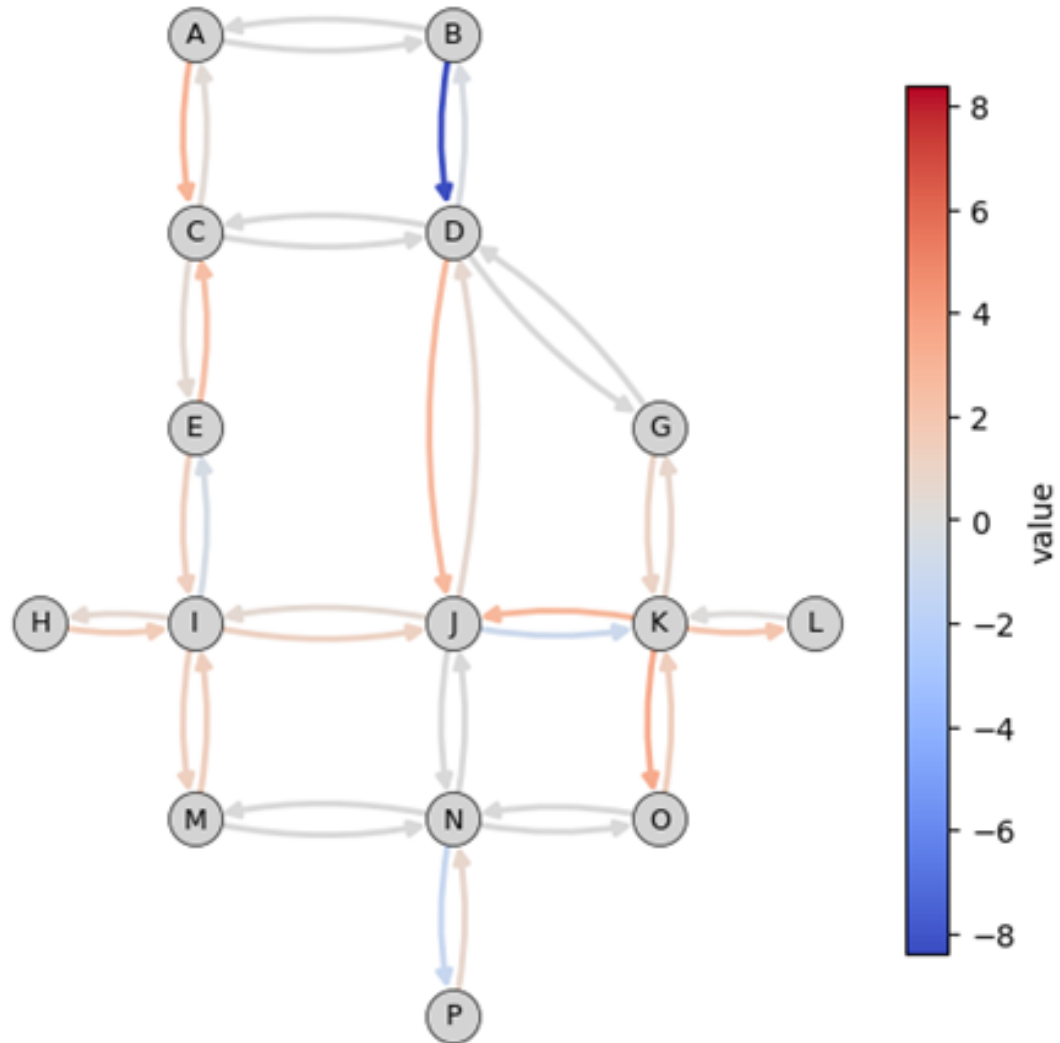


Figure B.15: Full-size version of Figure 6.3c: prediction error (prediction minus observation) on supervised edges at 11 February 2025, 08:45. Positive values (red) indicate overestimation; negative values (blue) indicate underestimation.

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